Probability & Statistics (P&S / LGP)

Fast-Track 10-Day Mastery Roadmap for Beginners • High-Yield Scoring Blueprint

The Beginner's Tactical Shift: Inverted Unit Sequence

Most beginners fail P&S by getting stuck on abstract probability definitions in Unit 1 & 3. The **LearnGeeks Tactical Method** starts with **Unit 2 (Curve Fitting & Regression)** and **Unit 5 (Hypothesis Testing)**. These units are 100% mechanical, rely on fixed standard normal equations and calculator workflows, and guarantee 28 to 35 direct marks before touching pure theory.

DAILY 4-HOUR HIGH-YIELD STUDY ROUTINE

Phase 1: Formula Setup (45 Mins) — FOUNDATION

Write down the governing formula and declare every variable explicitly (e.g., sample size n, mean $\bar{x}$, standard error $\sigma/\sqrt{n}$). Never jump to numbers before writing the general algebraic equation.

Phase 2: Model Drills (105 Mins) — EXECUTION

Solve 3 textbook model problems manually. Construct full summation tables ($\sum x$, $\sum y$, $\sum xy$, $\sum x^2$). Do not look at answers mid-step; complete table arithmetic cleanly with strict column headings.

Phase 3: Blind Verification (60 Mins) — TESTING

Pick 2 previous JNTUK exam questions. Solve without reference notes. Use Casio fx-991MS/ES **STAT/REG** mode to instantly check the final parameters (a, b, r, z).

Phase 4: Formula Sheet Log (30 Mins) — RETENTION

Write daily formulas onto your dedicated LearnGeeks 4-page cheat sheet. Identify where marks were dropped (e.g., misreading 1-tail vs 2-tail, wrong degrees of freedom).

JNTUK R20 UNIT BLUEPRINT & STARRED PRIORITY BREAKDOWN

UNIT	PRESCRIBED SYLLABUS & CORE TOPICS	STARRED LEARNGEEKS PRIORITIES (*)	NATURE OF UNIT
Unit 1	Skewness, Kurtosis, Population & Sample, Data Collection, Visualization.	**Skewness & Kurtosis*** **Data Collection*** **Data Visualization***	Semi-Numerical + Theory Concepts
Unit 2	Rank Correlation, Regression Lines, Correlation, Least Squares, Parabola & Power Curves.	**Rank Correlation*** **Regression Lines*** **Parabola & Power Curves***	100% Numerical (High Scoring)
Unit 3	Bayes' Theorem, Discrete & Continuous Random Variables, Standard Distributions.	**Bayes' Theorem*** **Normal/Poisson Models***	Analytical Probability + Distributions
Unit 4	Mean & Variance, Population Size, Point & Interval Estimation, Sampling Distributions.	**Mean & Variance*** **Confidence Estimation***	Formulas + Interval Estimations
Unit 5	Errors (Type I & II), Large & Small Samples, Null/Alt Hypothesis, Significance, 1-tail & 2-tail.	**Errors (Type I & II)*** **Null Hypothesis*** **One/Two Tail Z & t Tests***	Standard 5-Step Procedural Testing

Days 1 to 5: Curve Fitting, Regression & Hypothesis Testing

Day 1: Unit 2 • Method of Least Squares & Curve Fitting

STARRED FOCUS Unit 2 Core

Master setup of normal equations for: (1) Straight line $y = a + bx$: Normal equations $\sum y = na + b\sum x$ and $\sum xy = a\sum x + b\sum x^2$. (2) Second-degree parabola $y = a + bx + cx^2$ (3 equations: $\sum y$, $\sum xy$, $\sum x^2y$). (3) Power curve $y = ax^b$: Convert using logarithms $\log y = \log a + b \log x \Rightarrow Y = A + bX$.

• **Daily Target: Solve 2 straight line, 2 second-degree parabola, and 2 power curve problems without error.**

Day 2: Unit 2 • Correlation & Regression Analysis

STARRED FOCUS Unit 2 Full Mastery

(1) Karl Pearson's coefficient of correlation (r). (2) Spearman's Rank Correlation: Non-repeated ranks $\rho = 1 - [6\sum d^2 / (n(n^2 - 1))]$, and repeated ranks with correction factor $m(m^2 - 1)/12$. (3) Lines of Regression ($y - \bar{y} = b_{yx}(x - \bar{x})$ and $x - \bar{x} = b_{xy}(y - \bar{y})$). Properties: $r = \pm\sqrt{(b_{yx} \times b_{xy})}$.

• **Daily Target: Compute rank correlation for tied ranks (3 problems) + regression line prediction (2 problems).**

Day 3: Unit 5 • Hypothesis Testing Basics & Large Sample Means (Z-Test)

STARRED FOCUS Unit 5 Foundation

(1) Theory: Null (H_0) vs Alternative (H_1), Type I Error (α: reject true H_0), Type II Error (β: accept false H_0). (2) One-tailed vs Two-tailed tests; Level of significance ($1\% \rightarrow 2.58, 5\% \rightarrow 1.96, 10\% \rightarrow 1.645$). (3) Large sample tests ($n \geq 30$): Test for single mean $Z = (\bar{x} - \mu) / (\sigma/\sqrt{n})$, Difference of two means $Z = (\bar{x}_1 - \bar{x}_2) / \sqrt{[(s_1^2/n_1) + (s_2^2/n_2)]}$.

• **Daily Target: Master writing JNTUK's 5-step hypothesis format. Solve 4 large-sample numerical problems.**

Day 4: Unit 5 • Large Sample Proportions & Small Samples (t-Test)

CRITICAL NUMERICAL Unit 5 Completion

(1) Test for single proportion $Z = (p - P) / \sqrt{(PQ/n)}$ and difference of proportions $Z = (p_1 - p_2) / \sqrt{[p\hat{}q\hat{}(1/n_1 + 1/n_2)]}$. (2) Small sample Student's t-test ($n < 30$, σ unknown): Single mean $t = (\bar{x} - \mu) / (s/\sqrt{n})$ with $df = n - 1$. (3) Difference of means with pooled variance $S^2 = [(n_1-1)s_1^2 + (n_2-1)s_2^2] / (n_1 + n_2 - 2)$, and Paired t-test for dependent samples.

• **Daily Target: Solve 2 proportion problems, 2 independent t-tests, and 1 paired t-test problem.**

Day 5: Unit 3 • Bayes' Theorem & Random Variables

STARRED FOCUS Unit 3 Heavyweight

(1) Bayes' Theorem for partitions $B_1, B_2, \ldots, B_k$:

$P(B_i|A) = [\, P(B_i) \times P(A|B_i) \,] / [\, \sum_{j=1}^{k} P(B_j) \times P(A|B_j) \,]$ (Classic machine production & bolt defect problems).

(2) Discrete Random Variables: Find normalizing constant k using $\sum P(x) = 1$, Expectation $E(X) = \sum xP(x)$, and Variance $Var(X) = E(X^2) - [E(X)]^2$.

• **Daily Target: Solve 4 Bayes' theorem factory problems and 3 discrete distribution normalization problems.**

Days 6 to 10: Distributions, Estimation & Full Revision

Day 6: Unit 3 • Standard Distributions (Binomial, Poisson & Normal)

CORE PROBABILITY | Unit 3 Complete

(1) Binomial Distribution: $P(X = x) = {}^{n}C_{x}\, p^{x}\, q^{n-x}$, Mean $\mu = np$, Variance $\sigma^2 = npq$.

(2) Poisson Distribution: $P(X = x) = (e^{-\lambda} \times \lambda^{x}) / x!$, Mean = Variance = λ.

(3) Normal Distribution: Continuous density, conversion to Standard Normal $Z = (X - \mu) / \sigma$. Reading normal curve area tables, computing $P(a < X < b)$.

• Daily Target: Solve 2 Binomial fitting problems, 2 Poisson frequency problems, and 3 Normal curve area calculations.

Day 7: Unit 4 • Sampling Theory & Statistical Estimations

STARRED FOCUS | Unit 4 Mastery

(1) Definitions: Population vs Sample, Sampling Distribution, Standard Error (S.E.).

(2) Mean & Variance of sampling distribution of means: $\mu_{\bar{x}} = \mu$, $\sigma_{\bar{x}} = \sigma/\sqrt{n}$ (with FPC factor for finite populations).

(3) Estimations: Point vs Interval estimation. Confidence intervals for population mean: $\bar{x} \pm Z_{\alpha/2}(\sigma/\sqrt{n})$ at 95% and 99%. Sample size determination $n = [Z_{\alpha/2} \times \sigma / E]^2$.

• Daily Target: Solve 3 confidence interval problems (95% & 99%) and 2 sample size calculation problems.

Day 8: Unit 1 • Skewness, Kurtosis & Data Visualization

STARRED FOCUS | Unit 1 Theory & Tables

(1) Karl Pearson's Skewness: $S_k = (Mean - Mode) / \sigma = 3(Mean - Median) / \sigma$.

(2) Bowley's Quartile Skewness: $S_b = (Q_3 + Q_1 - 2Q_2) / (Q_3 - Q_1)$. Kurtosis: $\beta_2 = \mu_4 / \mu_2^2$ (Lepto, Meso, Platykurtic).

(3) Theory Topics: Primary vs Secondary data collection methods, Histograms, Frequency Polygons, Ogives, and Box plots.

• Daily Target: Solve 2 continuous series Skewness/Kurtosis table problems + write 3 standard descriptive theory answers.

Day 9: Full Formula Consolidation & Previous Paper (PYQ) Mock

EXAM SIMULATION | Complete Syllabus

(1) Write all normal equations, test statistics, and distribution formulas from memory onto a single sheet.

(2) Sit for a strict 3-hour timed mock test using a recent JNTUK R20 P&S question paper.

(3) Self-evaluate against university marking guidelines: identify incomplete step statements or missing calculator checks.

• Daily Target: Complete 1 full question paper (Answer 5 out of 8 complete questions with clean steps).

Day 10: Calculator Mastery, Weak-Spot Patching & Final Confidence

FINAL READINESS | Zero Panic Day

(1) Scientific Calculator Drill: Verify STAT mode for Mean/SD and REG mode for slope b, intercept a, and correlation r in under 45 seconds.

(2) Review starred concepts: Type I vs Type II errors, Bayes' Theorem, Rank correlation with ties, and 95% confidence intervals.

(3) Organize exam kit: Calculator (with fresh batteries), statistical normal distribution & t-tables, scale, pencils.

• Daily Target: Re-solve 5 previously failed problems; verify all calculator shortcuts; relax before exam day.

Casio fx-991MS / ES Plus Speed Operations for P&S

TIME SAVER

• **Linear Regression (Unit 2):** `MODE` → `REG` → `LIN`. Enter data: `x-val, y-val DT`. Press `SHIFT` → `S-VAR` to read `r`, A (intercept), and B (slope). Cross-check manual normal equations in 30 seconds.

• **Single Variable Statistics (Unit 1 & 4):** `MODE` → `STAT` → `1-VAR`. Enter frequencies. Press `SHIFT` → `1 (STAT)` to get $\bar{x}$, σ, and s instantly.

Exam Presentation Standard & Progress Tracker

Mandatory 5-Step Hypothesis Presentation (Unit 5 Evaluator Checklist)

In JNTUK R20 valuation, evaluators award step marks based strictly on this 5-step structure:

- **Step 1: Hypotheses** — State Null Hypothesis $H_0: \mu = \mu_0$ and Alternative Hypothesis $H_1: \mu \neq \mu_0$ (explicitly identify: Two-tailed or One-tailed).
- **Step 2: Level of Significance (α)** — Write $\alpha = 0.05$ (5%) or 0.01 (1%), and state Critical Table Value (e.g., $Z_{tab} = \pm 1.96$).
- **Step 3: Test Statistic Formula** — Write the general algebraic formula: $Z_{cal} = (\bar{x} - \mu) / (\sigma/\sqrt{n})$.
- **Step 4: Computation** — Substitute given sample parameters with intermediate values clearly calculated.
- **Step 5: Decision & Inference** — If $|Z_{cal}| < Z_{tab}$, write: *"Accept H_0. There is no significant difference at α level."* If $|Z_{cal}| \geq Z_{tab}$, write: *"Reject H_0."*

Unit 2 Curve Fitting Normal Equations Master Reference

CURVE TYPE	GOVERNING EQUATION	SYSTEM OF NORMAL EQUATIONS
Straight Line	$y = a + bx$	(1) $\sum y = na + b\sum x$ (2) $\sum xy = a\sum x + b\sum x^2$
Parabola (2nd Deg)	$y = a + bx + cx^2$	(1) $\sum y = na + b\sum x + c\sum x^2$ (2) $\sum xy = a\sum x + b\sum x^2 + c\sum x^3$ (3) $\sum x^2y = a\sum x^2 + b\sum x^3 + c\sum x^4$
Power Curve	$y = ax^b \Rightarrow Y = A + bX$	(1) $\sum Y = nA + b\sum X$ (2) $\sum XY = A\sum X + b\sum X^2$ (where $A = \log a$)

10-DAY STUDENT PROGRESS & COMPLETION TRACKER

DAY	TOPIC FOCUS	TARGET PROBLEMS	HOURS DONE	STATUS
Day 1	Curve Fitting (Line, Parabola, Power)	6 Problems (Least Squares)	[ ] / 4 hrs	[ ] Done
Day 2	Rank Correlation & Regression	5 Problems (Ties & Lines)	[ ] / 4 hrs	[ ] Done
Day 3	Hypothesis Basics & Large Means	4 Problems (Z-Tests)	[ ] / 4 hrs	[ ] Done
Day 4	Proportions & Student t-Test	5 Problems (t & Proportions)	[ ] / 4 hrs	[ ] Done
Day 5	Bayes' Theorem & Random Variables	7 Problems (Defects & k-value)	[ ] / 4 hrs	[ ] Done
Day 6	Binomial, Poisson & Normal	7 Problems (Distributions)	[ ] / 4 hrs	[ ] Done
Day 7	Sampling & Confidence Intervals	5 Problems (Estimation & n)	[ ] / 4 hrs	[ ] Done
Day 8	Skewness, Kurtosis & Visuals	2 Full Tables + 3 Theory Notes	[ ] / 4 hrs	[ ] Done
Day 9	Full Question Paper Simulation	1 Complete JNTUK PYQ Paper	[ ] / 4 hrs	[ ] Done
Day 10	Calculator Speed & Starred Review	Quick Checks & Weak Spots	[ ] / 4 hrs	[ ] Done